

LabCheck App Summary

Evidence source: repository code in src/, package.json, and README.md

What It Is

LabCheck is a React single-page web app for managing lab coursework across student, teacher, and administrator roles. It provides role-based dashboards, lab progress tracking, and admin management workflows using client-side state and seeded data.

Who It's For

Primary users are university students, teachers, and administrators managing lab-based courses; examples and validation rules target @polytechnic.am accounts.

What It Does

- Role-based login routes users to /student, /teacher, or /admin, with @polytechnic.am email validation.
- Student dashboard shows subjects, lab assignment cards, and a progress overview with absences, midterm eligibility, and final score (0-16).
- Teacher dashboard uses subject -> group -> subgroup navigation and supports two views: lab management and progress overview.
- Teacher lab manager adds, edits, and deletes labs by subject/group, including title, description, and difficulty (1-5), up to MAX_LABS=14.
- Teacher progress table records attendance and grades per lab, calculates absences/final status, and exports Not allowed students to .xlsx.
- Admin area includes pending approvals and in-page management for departments, groups, subjects, students, teachers, and assignments.

How It Works (Repo-Evidenced Architecture)

- Frontend stack: React 19 + React Router + Vite (BrowserRouter in src/main.jsx; routes in src/App.jsx).
- Auth/session: AuthContext keeps user in React state and navigates by role; persistence across refreshes is Not found in repo.
- Data flow: components read and mutate in-memory objects from src/data/*.js (labs, progress, groups, users).
- Services/API: network API clients, backend service layer, and database integration are Not found in repo.
- Signup path: SignupPage calls signup(), but a signup function in AuthContext is Not found in repo.

How To Run

- From the project root: npm install
- Start dev server: npm run dev
- Open the local URL printed by Vite (commonly http://localhost:5173)
- Production build command exists (npm run build); deployment instructions are Not found in repo.